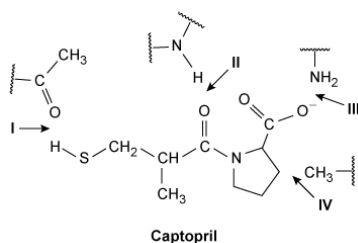


The structure of the antihypertensive drug captopril is shown below interacting with the active site of an enzyme.

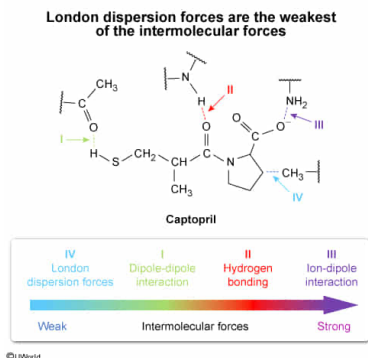


Which region depicts the weakest type of interaction?

- ☐ A. I
- ☐ B. II [1%]
- ☐ C. III [3%]
- ☒ D. IV [94%]

Correct
94% answered correctly

Explanation:



Covalent bonds are strong chemical bonds that arise when an electron pair is shared between two atoms, resulting in the formation of a molecule. In contrast, **noncovalent interactions** are weak **intermolecular attractions** that arise when the **dipoles** of two molecules interact. This **attractive** force occurs because the partial positive charge (δ^+) of one dipole is attracted to the partial negative charge (δ^-) of another dipole.

Noncovalent interactions include attractions between two **polar** molecules (**dipole-dipole interactions**), attractions between a polar and a **nonpolar** molecule (**dipole-induced dipole interactions**), and attractions between two nonpolar molecules (**induced dipole-induced dipole**, also known as **London dispersion forces**). These attractions are collectively called **van der Waals forces**, and the weakest of these are London dispersion forces.

Within a nonpolar molecule, the electron density is symmetrically distributed around the nucleus. However, when two nonpolar molecules are in close proximity, the electron density of one molecule can momentarily distort to form a slightly positive region that then induces a slightly negative region in the neighboring molecule. Consequently, temporary dipoles with weak mutual attraction are formed.

In region IV of the captopril-enzyme binding site, a nonpolar alkyl side chain is shown adjacent to a nonpolar methylene group ($-\text{CH}_2-$) in the ring structure. Therefore, the primary intermolecular interactions in region IV are London dispersion forces (the weakest type).

(Choice A) In region I, dipole-dipole interactions (a strong intermolecular interaction) occur between the δ^- of the $\text{C}=\text{O}$ polar bond and the δ^+ of the $\text{S}-\text{H}$ polar bond.

(Choice B) Region II involves **hydrogen bonding** (a special type of strong dipole-dipole interaction) resulting from the interaction between the oxygen from $\text{C}=\text{O}$ and the dipole from $\text{N}-\text{H}$.

(Choice C) In region III, the interaction between the negative charge on the O atom and the positive end of the polar $-\text{NH}_2$ group is an **ion-dipole interaction** (the strongest noncovalent intermolecular force).

Educational objective:

London dispersion forces, also called induced dipole-induced dipole interactions, are the weakest of the noncovalent van der Waals forces. These interactions occur between two molecules that are near enough to each other for distortions in the electron cloud to induce weak instantaneous dipoles.